

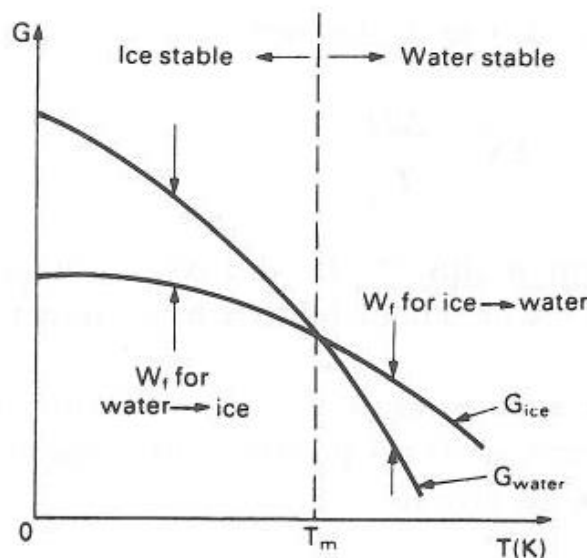
## 1. Basics: phase transformation

### 1. A touch of thermodynamics: driving force in phase transformation

System will tend towards thermodynamic equilibrium, at which the Gibbs energy  $G$ , is at a minimum. And  $\Delta G$  is Gibbs free energy. When  $\Delta G$  is negative, there is a driving force for system change.

For example, for pure water at  $0^\circ\text{C}$  ( $T_m$ ), there is no change in Gibbs energy when the solid becomes a liquid. We have neutral equilibrium.

For pure water at  $T < T_m$ ,  $G^s < G^l$ . There is a negative  $\Delta G$  if the liquid becomes a solid, leading to an unstable situation and a driving force for this change.



$$\Delta G = \Delta H - T\Delta S$$

For coarsening:

$$V_{\text{final}} = V_{\text{initial}}$$

$$\Delta G = (\Delta\gamma)_{\text{final}} - (\Delta\gamma)_{\text{initial}}$$

$\Delta G$  can be also quantified. For example, coarsening is a phenomenon where small precipitates tend to merge with bigger precipitates as there is a large driving force for this transformation.  $\Delta G$  can be calculated by the difference between total interfacial energy of the joined big particle and the total interfacial energy of the small particles.

### 2. Transformation rate influenced by temperature

$\Delta G$  can be expressed by the change of enthalpy ( $\Delta H$ ) and entropy ( $\Delta S$ ).

For a small departure from the melting point, the values of  $H$  and  $S$  can be considered independent of  $T$ . Therefore the driving force for transformation increase as the temperature decreases below the melting point.

$$\Delta G = \Delta H - T\Delta S \quad \therefore 0 = \Delta H - T_m \Delta S$$

$$\therefore \Delta S = \frac{\Delta H}{T_m} \quad \therefore \Delta G = \Delta H \left( \frac{T_m - T}{T_m} \right)$$

$$\Delta G = \Delta H - T\Delta S$$

$$0 = \Delta H - T_m \Delta S$$

$$\Delta S = \frac{\Delta H}{T_m}$$



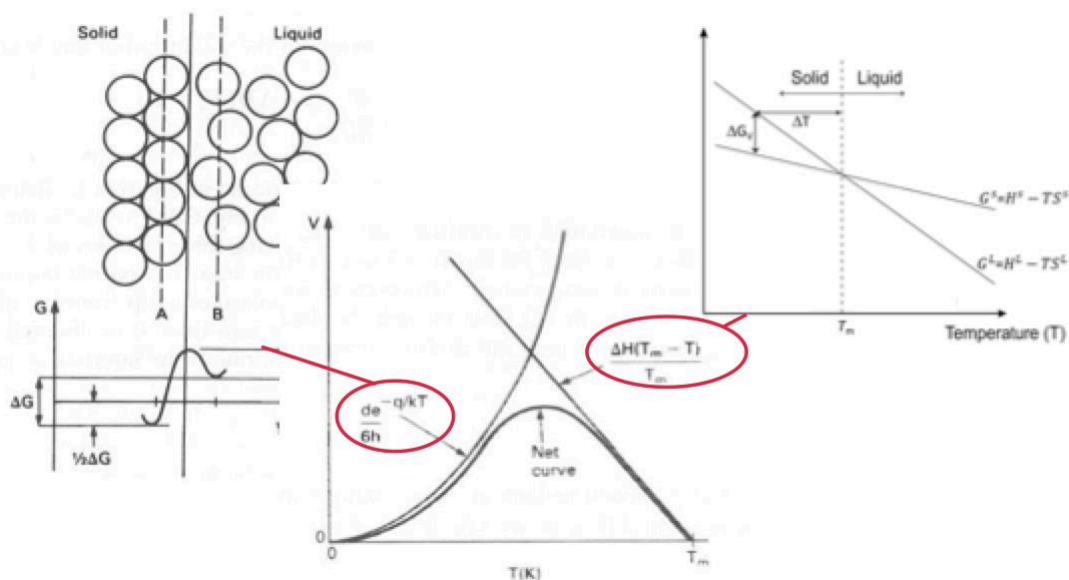
$$\Delta G = \Delta H - T \frac{\Delta H}{T_m}$$

Undercooling

$$\Delta G = \Delta H \left( \frac{T_m - T}{T_m} \right)$$

On the other hand, the probability that an atom will jump from the liquid to the solid decrease (the mobility of the atoms) as the temperature decreases.

Therefore as shown below the transformation rate increase as the temperature decreases due to the increase driving force. Then the rate decreases as the low temperature reduces atomic motion, giving an optimum temperature where transformation rate is maximal.

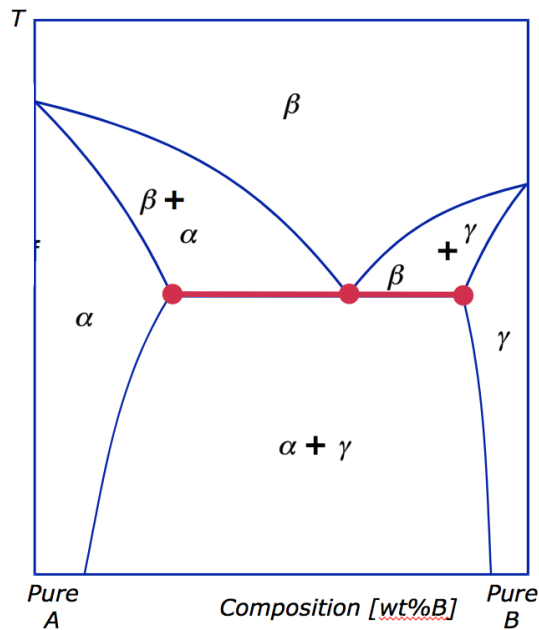


### 3. Diffusion controlled phase transformation

Diffusion is mass transport by atomic motion. In many cases, a phase transformation requires the diffusion of solute away from a growing phase, such as

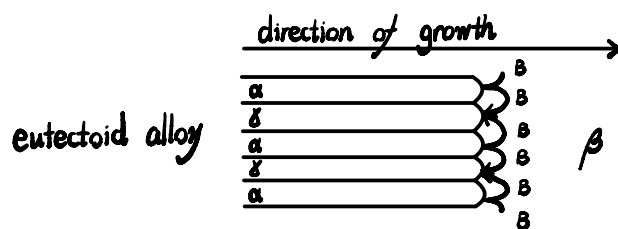
Eutectic growth:  $L \rightarrow \alpha + \beta$

Eutectoid growth:  $\beta \rightarrow \alpha + \gamma$



For example, for eutectoid growth,  $\alpha$  is a phase containing a large amount of A and small amount of B; while  $\gamma$  is a phase containing large amount of B and small amount of A. In this case the speed of reaction is determined by the rate of transport of solute from that phase. The laminar geometry of the phases is also formed because this is the fastest way for B to transform from phase  $\alpha$  to phase  $\gamma$ .

Sketch what microstructure of the eutectoid phase is expected during the phase transformation (including the B atoms moving direction).



#### 4. Nucleation of a new phase

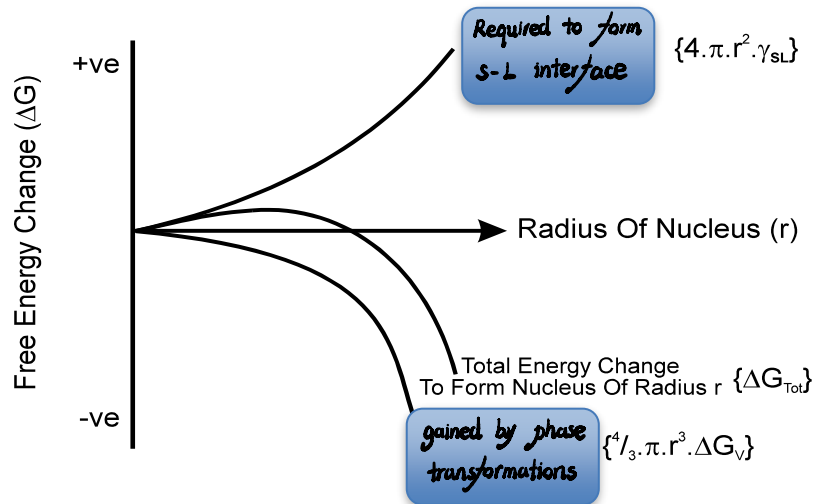
*Homogenous nucleation theory*

In this theory, the Gibbs free energy is influenced by the energy required to form a new interface (S-L interface) and the energy gained by phase transformation from liquid to solid.

Therefore near to the  $T_m$ , clusters of atoms with the solid structure (nuclei) will appear in the liquid but these nuclei will only grow if increase their size will result in a lowering of Gibbs's free energy. So there is a critical radius above which all nuclei will grow corresponds to the maxima in the total free energy.

Mark each part of the equation below in the diagram. What energy do they express.

$$\Delta G_{Tot} = \left( -\frac{4}{3}\pi r^3 \Delta G_V \right) + (4\pi r^2 \gamma_{SL})$$

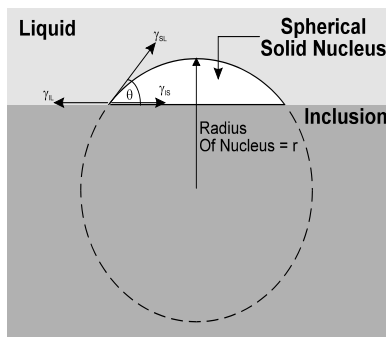


The critical radius is inversely proportional to the current level of undercooling and it is extremely improbable that more atoms than needed for a 1nm sphere in the liquid will spontaneously form a solid arrangement. This translates into a required undercooling of  $\Delta T \sim 100K$ .

$$r_{crit} = \frac{2 \cdot \gamma_{SL}}{\Delta G_V}$$

#### Heterogeneous nucleation theory

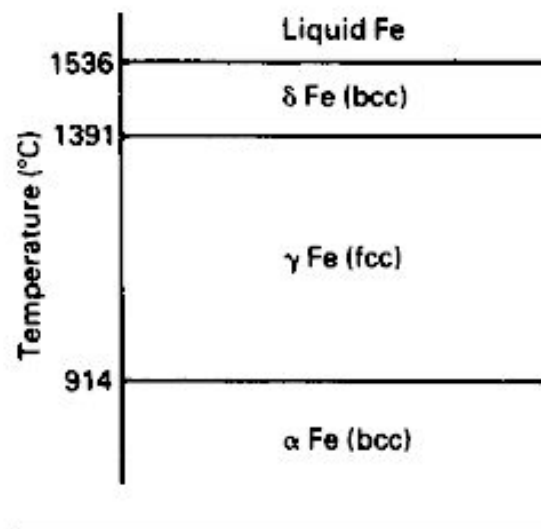
In reality nucleation occurs on the surface of crucible / inclusion. Here there already exists a S-L interface so when a nuclei forms the new S-L interface has less interface, leading to less required energy to create a new surface.



Nucleants such as  $TiB_2$  or  $Al_3Ti$  to help the heterogeneous nucleation.

Additionally, a solid phase contains many high energy defects, e.g. dislocations, grain boundaries. These regions of higher energy are good heterogeneous nucleation sites. Homogeneous nucleation is very rare in a solid phase.

## 5. Diffusive vs. displacive phase transformations in pure iron



Iron (Fe) has polymorphism. γ-Fe and α-Fe are engineering related phases (more details see Topic Steel).

$\gamma \rightarrow \alpha$  diffusive transformation

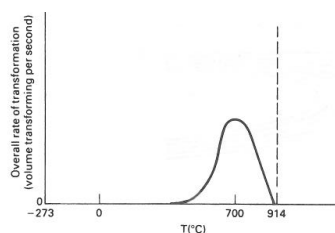
Phase transformations are nucleation and growth processes.

When we have a larger undercooling, the critical radius will

smaller, the driving force/ Gibbs free energy will be larger, but the thermal energy for the nucleation will be smaller, the mobility of the atom will be reduced.

So the phase transformation rate is proportional to the numbers of α nuclei and the growth rate of α-γ interface.

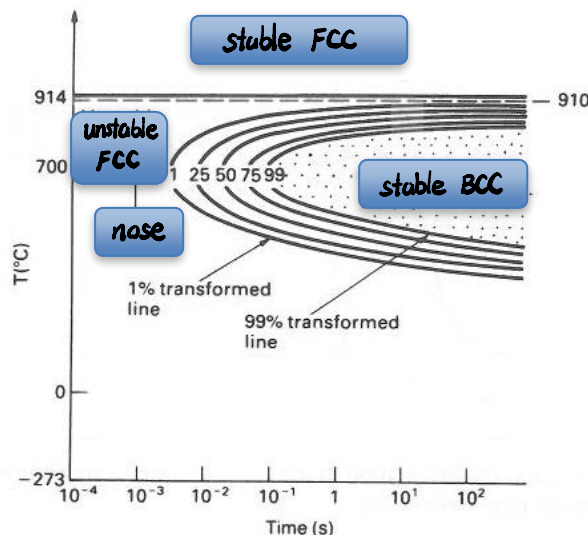
$$\text{Transformation rate (volume s}^{-1}\text{)} \propto \text{Number of } \alpha \text{ nuclei} \times \text{Growth rate of } \alpha\text{-}\gamma \text{ interface}$$



Time - Temperature - Transformation (TTT) curves can be used to indicate at certain temperature and certain time, what the phase transformation status is.

Mark stable fcc iron, unstable fcc iron, stable bcc iron, “nose” in the curve. What does “25” on the curve mean?

25% unstable FCC iron  $\rightarrow$  stable BCC iron



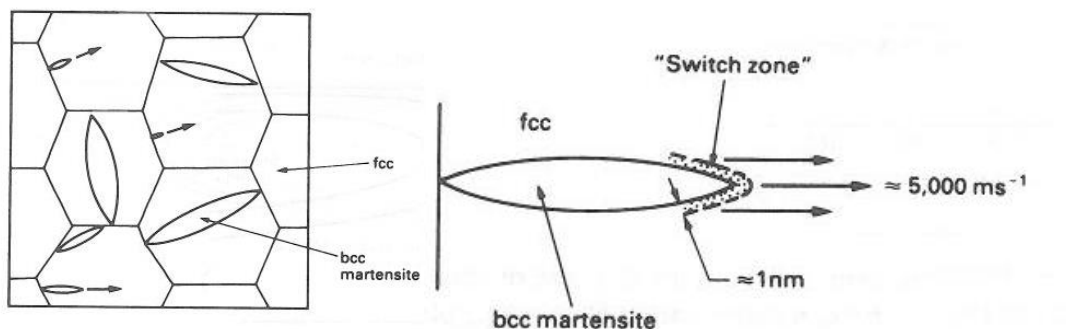
$\gamma \rightarrow \alpha$  displacive transformation (Martensitic transformation)

In Fe, the system has another way to transform which is so called

displacive transformation.

In this transformation, small lenses of BCC-Fe form at

grain boundaries. These propagate across grains at the speed of sound. There is a “switch zone” in which atomic bonds are broken and remade to change from FCC to BCC and this transformation is similar to dislocation motion breaking and remaking bonds.



As lenses grow, they distort the lattice planes and some of the Gibbs free energy ( $\Delta G$ ) is absorbed as strain energy, therefore martensite-iron is very hard (more see the topic of Steel)

If all the available  $\Delta G$  is absorbed as strain energy, the transformation stops.

Therefore, there must be enough  $\Delta G$  to start the displacive transformation (i.e. provide the strain energy). The volume fraction transformed then

increased with decreasing temperature (increasing  $\Delta G$  available to accommodate the strain energy).

In the Time - Temperature - Transformation curves below, process 1 gives a diffusive transformation and process 2 gives a displacive transformation. The image below presents the microstructure of BCC martensite growing. The arrow 1 points at FCC phase, and arrow 2 points at BCC martensite phase.

